Same search in both arms
extract128 :: $\langle$lift, Feistel, $P\rangle$ on one target $T = Q$
folded index, 14,940,300 states, same order (key = hash of Q)
per state: 83 shifts $\times$ 2 stored roots R_k; first lifted relation

Baseline cS: S3 loop (P = compact)
per candidate: $x(\phi^s(R_k))$ to polynomial basis,
batched S3 solve $\Rightarrow \{x(T+\phi^s R_k),\, x(T-\phi^s R_k)\}$
archived: 221.4 ns per probe (run 39214027289)

Candidate cF: frame loop (P = frame)
$T_s = \phi^{-s}(T)$ tabulated once per target
$x(T \pm \phi^s R_k) = \phi^s\left(x(T_s \pm R_k)\right)$ by point addition,
S3 order resolved only at a root-table hit
rank stage: 141.5 ns per probe (same code)

Gates (every point): one scalar (oracle replay, and tuned rho rT), one relation (indices, x codes, intermediates), one probe count, root calls, relation checks across cS, cF, cFaa, cFX (and kP on its unit); B's index digests; Q0 = archived Feistel identity (1,987,852 probes).
Controls: A/A cF / cFaa in [0.97, 1.03]; planted cFX = frame loop + the S3 loop redone and compared on every state, online lower bound > 1.
Primary: R_{on} = mean online(cS) / mean online(cF), 64 new points (onlineframe-v1), paired bootstrap; improvement iff 95% lower bound > 1.
Secondary: end-to-end cold with one shared rank stage K (kP, seed 701) on both sides; pipeline improvement only if its lower bound > 1.